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Managing Underperformance Risk in Project Portfolio Selection

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We consider a project selection problem where each project has an uncertain return with partially characterized probability distribution. The decision maker selects a feasible subset of projects so that the risk of the portfolio return not meeting a specified target is minimized. To model and evaluate this risk, we propose and justify a general performance measure, the underperformance riskiness index (URI). We define a special case of the URI, the entropic underperformance riskiness index (EURI), for the project selection problem. We minimize the EURI of the project portfolio, which is the reciprocal of the absolute risk aversion (ARA) of an ambiguity-averse individual with constant ARA who is indifferent between the target return with certainty and the uncertain portfolio return. The EURI extends the riskiness index of Aumann and Serrano (2008) by incorporating the target and distributional ambiguity, and controls the underperformance probability (UP) for any target level. Our model includes correlation and interaction effects such as synergies. Since the model is a discrete nonlinear optimization problem, we derive the optimal solution using Benders decomposition techniques. We show that computationally efficient solution of the model is possible. Furthermore, the project portfolios generated by minimizing the underperformance risk are more than competitive in achieving the target with those found by benchmark approaches, including maximization of expected return, minimization of UP, mean-variance analysis, and maximization of Roy's safety-first ratio (1952). When there is only a single constraint for the budget, we describe a heuristic which routinely provides project portfolios with near-optimal underperformance risk.

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1. Introduction

One-fifth of the world's economic activity, or \$12 trillion annually, is organized as a project (Project Management Institute 2013). In the numerous organizations that own and manage projects, a problem of central importance is *selecting* which of the organization's available projects to accept and run. This decision is typically made either by a Project Management Office, or by a *project selection committee* composed of senior managers with experience as project managers. Statistical and anecdotal evidence suggests that doing the right projects is a big factor in doing projects right (Cooper et al. 2000). Indeed, well-chosen projects are typically relatively easy to manage. Whereas, poorly selected projects are often dysfunctional and may compromise other projects by absorbing their resources.

Baker and Freeland (1975) provide an early review of the project selection literature. Chien (2002) provides an updated review, and observes that most of the literature considers project selection decisions that are made *sequentially*, i.e., one project at a time (iSixSigma.com 2010). Chien (2002) emphasizes that this approach does not generate optimal project portfolios. Heidenberger and Stummer (2003) also review the project selection literature, focusing on applications of operations research models.

We briefly review the literature of sequential project selection. In many organizations, available projects are initially screened according to various criteria such as their payback period and risk characteristics. Those projects that pass the initial screening are subjected to a more detailed scoring and/or ranking analysis. Nelson (1986) provides a scoring model for selecting flexible manufacturing system

projects. Lu et al. (1999) study sequential project evaluation for dynamically arriving projects. Henriksen and Traynor (2002) describe a scoring model for selecting research and development projects. Meade and Presley (2002) describe a multifactor ranking process that takes into account an organization's strategic objectives and culture. Poh et al. (2002) provide a comparison of project selection methods that use the analytic hierarchy process. All these works use sequential project selection.

However, a *project portfolio optimization* approach that simultaneously makes accept or reject decisions for all the available projects offers several significant advantages over sequential project selection (Chien 2002, Kooragamage 2010). First, it more accurately models overall project portfolio issues such as risk. Second, it more effectively uses the available resources. Third, it enables the modeling of correlation between uncertain project returns. Fourth, where there are interaction effects such as synergies between the projects, they can also be modeled effectively. Finally, the project portfolio approach enables optimization of an overall objective for project selection. For these reasons, our work considers project selection as a portfolio optimization problem.

A basic mathematical model that supports binary project selection decisions is the deterministic zero-one knapsack problem (Kellerer et al. 2004) and its extensions. Hall et al. (1992) develop a zero-one knapsack model with additional constraints, and apply it to project funding decisions at the National Cancer Institute. Generalizations of the knapsack model (Fox et al. 1984, Dickinson et al. 2001) model interaction effects that are limited to those arising from synergistic returns between pairs of projects. Keisler (2005) identifies synergies among larger groups of projects. All these works consider the project selection problem as having known deterministic returns and other data.

However, our work recognizes that, in many practical project selection problems, there is a high level of uncertainty in the return from each project (Pohl and Mihaljek 1992). This uncertainty may arise either from uncertainty about its technical success, or from uncertainty about its commercial success (MacMillan and McGrath 2002). Technical uncertainty arises from uncertain outcomes in research and development (e.g., for new products), in prototype testing (e.g., for safety testing of automobiles), and in regulatory approval (e.g., for new pharmaceutical products). Commercial uncertainty arises from randomness in time to market (e.g., for seasonal products such as fashion items and toys), in the introduction of competitors' products (e.g., for consumer electronics), and, in general, economic factors (e.g., a recession). Henig (1990) models a project selection problem with uncertain returns as a stochastic knapsack problem, with the objective of maximizing the probability of attaining a given target for total return. Kleywegt and Papastavrou (1998) extend the literature of the stochastic knapsack problem by considering items, i.e., projects, which arrive according to a Poisson

process, with resource requirements and rewards that are uncertain. They provide optimal recursive and closed-form solutions for various cases. Our work considers a variety of general constraints, possibly including a knapsack budget constraint.

Other optimization approaches besides those based on the knapsack problem are also studied. Loch and Kavadias (2002) consider the problem of dynamically adjusting investment levels in programs or projects, resulting in higher or lower returns, using the concept of marginal returns to investment. Carlsson et al. (2007) develop a fuzzy mixed-integer programming model, incorporating the value of managerial flexibility using real options. Their problem includes only a single constraint for the available budget. Consideration of these problem extensions is potentially valuable, however our work focuses on the classical binary choice project selection problem, with uncertainty and interaction effects.

Finally, several works develop customized project selection procedures for special applications. These include Cooper (1985) for new product selection, Nelson (1986) for manufacturing modernization projects, Oral et al. (2001) for project selection problems with competing interests among multiple stakeholders, and Kolisch and Meyer (2006) for pharmaceutical research applications. Our work, however, is motivated by generic applications without industry-specific features.

In the project selection literature, there are two important issues, which have received insufficient attention. The first issue is the unavailability of full distributional information about the uncertain returns. Because of the unique, non-recurring characteristics of projects (Newell et al. 2006), probability distributions for returns may not be known. For the project selection problem, classical robust optimization can be applied to maximize the worst-case total return, when the returns of projects are described by an uncertainty set (Soyster 1973, Ben-Tal and Nemirovski 1998). Kouvelis and Yu (1997) consider similar uncertainty sets, which can be represented by a convex hull of scenarios. Bertsimas and Sim (2004) introduce an adjustable polyhedral uncertainty set, where the decision maker specifies an ambiguity preference by defining a "budget of uncertainty." These approaches do not make use of distributional information other than bounds. We consider the project selection problem under partially characterized distributions, or *distributional ambiguity*, about project returns.

The second issue is the objective function, which maps the uncertain return to a decision criterion. Ideally, the objective function should be computable in polynomial time. Unfortunately, even in the absence of ambiguity, determining the distribution of the total portfolio return, which involves the evaluation of a weighted sum of random factors, is a computationally intractable problem (Khachiyan 1989, Nemirovski and Shapiro 2006). Moreover, the intractability issue may be exacerbated by the presence of distributional ambiguity, where the evaluation

of the objective function may require optimization over the distributional uncertainty set. A tractability requirement disqualifies widely used criteria such as many coherent and convex risk measures (Artzner et al. 1999, Föllmer and Schied 2004), including the conditional value-at-risk (CVaR) measure of Rockafellar and Uryasev (2000) and the optimized certainty equivalent of Ben-Tal and Teboulle (2007). Similar concerns disqualify expected utility under general utility functions, and cumulative prospect theory (Tversky and Kahneman 1992). The only utility function that can be tractably evaluated is the exponential utility function. However, a problem with this approach is how to elicit the risk-aversion parameter of the decision maker, since this may be difficult to specify objectively. Moreover, behavioral paradoxes, such as those of Allais (1953) and Ellsberg (1961), suggest weaknesses of the expected utility paradigm in predicting human preferences under uncertainty (Kahneman and Tversky 1979).

Worst-case expected return is appealing due to its intuitiveness and tractability; however, although it considers ambiguity aversion, it ignores risk aversion. See Knight (1921) and Hsu et al. (2005) for the distinction between risk and ambiguity. Mean-variance approaches, including Markowitz's (1959) and Roy's safety-first ratio (1952), are computable but contradict the principle of monotonicity. That is, a portfolio with returns that outperform those in another portfolio in all scenarios may be ranked inferior. Hence, these approaches are also unsuitable. We present a closely related approach that resolves these issues.

Our selection criterion is related to the concept of target-based choice (Simon 1955), which argues that the main goal of most firms is not maximizing return but rather attaining a target return. Several descriptive studies of the risk behaviors of real-world managers (Lanzillotti 1958; Mao 1970; Payne et al. 1980, 1981) confirm that aspiration levels drive decision making. Hence we allow the decision maker to stipulate a desired aspiration level or target return. We propose a general performance measure, the underperformance riskiness index (URI), to evaluate the risk. We further define a special case, the EURI, for the project selection problem. Recall that, in decision theory, given any utility function u , the absolute risk aversion (ARA) is defined by $(-u''(x)/u'(x))$, which represents the degree of risk aversion at the payoff level x . The EURI is the reciprocal of the ARA of an ambiguity-averse individual with constant ARA who is indifferent between the target return with certainty and the uncertain portfolio return. It extends the riskiness index of Aumann and Serrano (2008) by incorporating the target and distributional ambiguity, and controls the UP for any target level. Intuitively, by minimizing the EURI, our model identifies the least risky project portfolio that meets the given target.

We briefly summarize our contributions to the literature as follows:

1. We model interactions and correlation among project returns, which are distinct and important issues in the

project selection problem. We introduce the concept of a “project bundle” to formulate project interactions, and suggest a linear factor-based model to model correlation.

2. We describe uncertainties using a general distributional uncertainty set. This set contains conic constraints, and hence can incorporate moment and Kullback-Leibler divergence information (Kullback and Leibler 1951).

3. We propose a new performance measure, the URI. This enables us to consider the target return, which is an important factor in real decision making.

4. We provide a Benders decomposition algorithm to derive exact solutions efficiently. When there is only a single constraint for budget, we describe a greedy heuristic by which decision makers can easily find a close to optimal solution.

5. Our computational studies show that the projects selected by minimizing the underperformance risk are more than competitive in achieving the target with those found by several classical benchmark approaches, and are also more robust in performance.

This paper is organized as follows. Section 2 defines the problem, describes the interaction and uncertainty structure, and presents our model. Section 3 introduces the URI, and justifies using the EURI for the project selection problem. Section 4 describes an overall solution procedure. Section 5 contains a discussion of the solvability of the problem, and a simple heuristic for the special case with a single budget constraint. Section 6 compares the performance of the solutions found by the EURI against several classical benchmarks. Section 7 provides a conclusion, some comments about implementability issues, and suggestions for future research. Proofs appear in an appendix.

2. Model Formulation

Section 2.1 provides the notation and defines the project selection problem. Section 2.2 describes the modeling of interaction effects, uncertainty, and correlation. Section 2.3 describes how we model distributional ambiguity.

2.1. Notation and Problem Definition

Consider a set $\mathcal{N} = \{1, \dots, N\}$ of available projects. All projects are initially available. We use “ $\sim$ ” to denote uncertain quantities, for example, $\tilde{r}_n$ denotes the uncertain return of project n . Let $y_n = 1$ if project n is selected, and $y_n = 0$ otherwise. We use boldface notation for matrices and vectors, e.g., $\mathbf{y} = (y_1, \dots, y_N)$.

We denote the total uncertain return of the selected projects by $\tilde{\pi}(\mathbf{y})$. The problem faced by the decision maker is

$$\max \tilde{\pi}(\mathbf{y}) \quad (1)$$

$$\text{s.t. } \mathbf{A}\mathbf{y} \leq \mathbf{b}, \quad (2)$$

$$\mathbf{y} \in \{0, 1\}^N. \quad (3)$$

Objective (1) maximizes the total uncertain return. Constraints (2) require that the selection of projects meets various deterministic restrictions, where $\mathbf{A}$ is a constraint matrix and $\mathbf{b}$ is a vector of resources. Constraint (3) ensures that each project is either accepted in full or rejected. We let $\mathcal{Y} = \{\mathbf{y}: \mathbf{A}\mathbf{y} \leq \mathbf{b}, \mathbf{y} \in \{0, 1\}^N\}$. If there are no interactions between the returns of projects, then the total uncertain return of the selected projects is $\tilde{\pi}(\mathbf{y}) = \sum_{n=1}^N \tilde{r}_n y_n$. The next section relaxes this assumption.

2.2. Interactions, Uncertainty, and Correlation

Interactions among projects are common in practice. When several projects are implemented together, the total return can be greater or smaller than the sum of those projects' individual returns. For example, the implementation of two information technology (IT) projects may create additional value through integration (Cho 2010). Alternatively, two new product development projects that generate similar products in a small consumer market may compromise each other's profitability (Huber et al. 1982). Yet, in the project selection literature, few papers adequately address this phenomenon. This is apparently due to the complexity that is introduced by modeling it. One approach (Fox et al. 1984) is to use the cross product of the binary decision variables. However, this formulation defines a nonlinear combinatorial optimization problem with limited solvability, and moreover, returns for interactions among more than two projects are not considered.

We consider interactions among the uncertain returns of a subset of projects with any cardinality. We define a *project bundle* $\beta \subseteq \mathcal{N}$ to consist of any number of projects that may have an interaction effect. By project screening, the project selection committee can identify all important interaction effects among projects. Let $\mathcal{E}$ denote the basic set, which contains all bundles identified by the project selection committee as having significant interactions. In practice, the information available to the committee may be incomplete. For this reason, the return of some project bundles, for example, those containing two or more project bundles in the set $\mathcal{E}$, may not be well defined. This motivates the following definition.

DEFINITION 1. The *project bundle set*, $\mathcal{E}$, is the minimum set with the following properties:

1. $\{i\} \in \mathcal{E}$ for $i \in \mathcal{N}$;
2. $\mathcal{E} \subseteq \mathcal{E}$; and
3. for $\beta_1, \beta_2 \in \mathcal{E}$, if $\beta_1 \cap \beta_2 \neq \emptyset$, then $\beta_1 \cup \beta_2 \in \mathcal{E}$.

The first property establishes consistency with the simple case without interaction effects. From the second property, the project bundle set $\mathcal{E}$ includes all the significant interactions identified by the project selection committee. The third property shows that, for any two nondisjoint project bundles in $\mathcal{E}$, their union is a project bundle in $\mathcal{E}$. This definition of project bundles generalizes earlier studies (Fox et al. 1984, Dickinson et al. 2001) by considering interactions among more than two projects.

An example illustrates the project bundle set $\mathcal{E}$. Consider three available projects, $\mathcal{N} = \{1, 2, 3\}$. The project selection committee identifies a synergistic effect when projects 1 and 2 are selected, and when projects 2 and 3 are selected. Therefore $\{1, 2\}$ and $\{2, 3\}$ are project bundles in the basic set $\mathcal{E}$ and $\mathcal{E} = \{\{1, 2\}, \{2, 3\}\} \subseteq \mathcal{E}$. Further, from Property 3 of Definition 1, $\{1, 2, 3\}$ is also an element in $\mathcal{E}$. Hence the project bundle set is $\mathcal{E} = \{\{1\}, \{2\}, \{3\}, \{1, 2\}, \{2, 3\}, \{1, 2, 3\}\}$.

It follows that the total uncertain return corresponding to selection $\mathbf{y}$ is given by

$$\tilde{\pi}(\mathbf{y}) = \sum_{\beta \in \mathcal{E}} \tilde{r}_\beta \Gamma(\beta, \mathbf{y}),$$

where $\tilde{r}_\beta$ represents the uncertain return from selecting β , and $\Gamma(\beta, \mathbf{y})$ is an indicator function for bundle β from the selection $\mathbf{y}$ defined by

$$\Gamma(\beta, \mathbf{y}) = \begin{cases} 1 & \text{if } y_n = 1 \text{ for } n \in \beta; \nexists \beta' \in \mathcal{E}, \beta' \supset \beta, \\ & \text{such that } y_n = 1 \text{ for } n \in \beta'; \\ 0 & \text{otherwise.} \end{cases}$$

Thus, if $\Gamma(\beta, \mathbf{y}) = 1$, project bundle β is a maximal subset of $\mathcal{N}$ under selection $\mathbf{y}$. In general, it is unrealistic for the project selection committee to identify interaction effects among all subsets of projects. Instead, they typically focus on a small number of significant interactions. Hence $|\mathcal{E}|$ is not exponentially large in practice. Keisler (2005) describes the process used by managers to estimate $\tilde{r}_\beta$.

Our model of uncertainty is defined by a state-space Ω and a σ -algebra $\mathcal{F}$ of events in Ω . Consistent with the above discussion of the sources of uncertainty, we assume that the uncertain return of each project bundle is an affine function of bounded, independently distributed factors $\tilde{\mathbf{z}} = (\tilde{z}_1, \dots, \tilde{z}_K)$, i.e.,

$$\tilde{r}_\beta = r_\beta^0 + \sum_{k=1}^K r_\beta^k \tilde{z}_k,$$

where the factor coefficients $r_\beta^0, \dots, r_\beta^K$ are known. We thus describe the correlation effect using a factor-based model instead of the typical covariance matrix. Our reasons for doing so are (a) estimating the covariance matrix from real data is difficult, in general, especially for new or unique projects, (b) the linear factor-based model is motivated by practice since the performance of different project bundles depends on common factors in many cases, and (c) the linear factor-based model preserves the linear model structure and hence reduces the complexity of the solution process. In situations where historical data for similar projects is available, various statistical tools such as principal component analysis and linear regression can be used to calibrate the factors from the data. Even for a unique project, the factor-based model makes estimating the returns easier. Nevertheless, the linear factor-based model has some limitations. It does not model nonlinearity, and the coefficients

of the return factor matrix are assumed to be deterministic. The latter compromise is needed for tractability.

The returns of two selected projects are correlated if they have one or more factors with nonzero factor coefficients in common. The total uncertain return from project selection $\mathbf{y}$ is

$$\tilde{\pi}(\mathbf{y}) = \sum_{\beta \in \mathcal{E}} \tilde{r}_\beta \Gamma(\beta, \mathbf{y}) = \sum_{\beta \in \mathcal{E}} r_\beta^0 \Gamma(\beta, \mathbf{y}) + \sum_{k=1}^K \left(\sum_{\beta \in \mathcal{E}} r_\beta^k \Gamma(\beta, \mathbf{y}) \right) \tilde{z}_k.$$

Further, we denote by $\mathcal{V}$ the space spanned by all possible project returns, $\tilde{v}: \Omega \rightarrow \mathbb{R}$, as follows:

$$\mathcal{V} = \left\{ \tilde{v}: \exists (v^0, \mathbf{v}) \in \mathbb{R}^{K+1} \text{ such that } \tilde{v}(\omega) = v^0 + \sum_{k=1}^K v^k \tilde{z}_k(\omega), \omega \in \Omega \right\}.$$

REMARK 1. We clarify the difference between correlation and interaction. For example, consider three available projects A, B, and C. Their returns depend on two independent uncertain factors $\tilde{z}_1, \tilde{z}_2$ by $\tilde{r}_A = \tilde{r}_B = \tilde{z}_1, \tilde{r}_C = \tilde{z}_2$. The interaction effect exists only between A and C such that $r_{AC} = 3(\tilde{z}_1 + \tilde{z}_2)$. Thus A and B are correlated, since they depend on the same uncertain factor $\tilde{z}_1$, but have no interactions, i.e., $\tilde{r}_{AB} = \tilde{r}_A + \tilde{r}_B = 2\tilde{z}_1$; whereas, A and C have interactions, i.e., $r_{AC} \neq \tilde{r}_A + \tilde{r}_C$, but are uncorrelated; and B and C are neither correlated nor do they have any interactions. Suppose $\tilde{z}_1$ can be either 10 or 20 with equal probability, $\tilde{z}_2$ is always 0, and we need to choose two out of the three projects. Given the interaction effect, it is optimal to choose A and C. Nevertheless, if we fail to consider the interaction effect in the model, we would select A and B, which is clearly not optimal.

2.3. Modeling Risk and Ambiguity

Since projects possess aspects of uniqueness (Project Management Institute 2004), there is frequently a lack of historical information about the probability distributions of their returns. Even with the availability of historical records, it is not always possible to determine exact distributions. Therefore we do not assume knowledge of the exact probability distribution on $\mathcal{F}$. Instead, we allow ambiguity and assume that the true distribution, $\mathbb{P}$, lies in a distributional uncertainty set denoted by $\mathcal{P}$.

Specifically, using the independence property of the uncertain factors, we define our distributional uncertainty set as $\mathcal{P} = \mathcal{P}_1 \times \cdots \times \mathcal{P}_K$, where

$$\mathcal{P}_k = \left\{ \mathbb{P}: \mathbb{P}(\tilde{z}_k \in \{z_{k1}, \dots, z_{kL_k}\}) = 1, \mathbb{E}_{\mathbb{P}}(\mathbf{g}_k(\tilde{z}_k)) \in \mathcal{C}_k \right\},$$

$$k = 1, \dots, K, \quad (4)$$

is the set of all possible probability distributions for the uncertain factor $\tilde{z}_k$. In the Definition (4), $z_{kl}, l = 1, \dots, L_k$

are all possible realizations of the uncertain factor $\tilde{z}_k$, $\mathbf{g}_k$ is a mapping from $\mathbb{R} \rightarrow \mathbb{R}^{M_k}$, and $\mathcal{C}_k$ is a convex set defined as

$$\mathcal{C}_k = \left\{ \mathbf{s} \in \mathbb{R}^{M_k}: \exists \mathbf{h} \in \mathbb{R}^{Q_k} \text{ such that } G_k \mathbf{s} + H_k \mathbf{h} \preceq_{\mathcal{H}_k} \mathbf{d}_k \right\},$$

for $G_k \in \mathbb{R}^{P_k \times M_k}, H_k \in \mathbb{R}^{P_k \times Q_k}, \mathbf{d}_k \in \mathbb{R}^{P_k}$, where $\mathcal{H}_k$ is a regular cone, i.e., a closed, convex, and pointed cone with nonempty interior.

The first restriction of our distributional uncertainty set defined in Equation (4) is motivated by *scenario building*, an approach that is commonly used in practice to analyze project payoffs (Hoffman 1993). For robust optimization, a similar concept has also been employed by Thiele (2008) and Ben-Tal et al. (2013). In particular, $z_{kl}, l = 1, \dots, L_k$, represents the realization of the uncertain factor at each potential scenario. The second restriction in Equation (4) specifies the probability distribution by constraining the general estimation, $g_k(\tilde{z}_k)$, which may include any descriptive statistics such as mean, variance, within a confidence set, $\mathcal{C}_k$. The confidence set is defined via an auxiliary vector $\mathbf{h}$. This vector allows us to model a rich variety of distributional information of $\tilde{z}_k$ in a unified way. Under the independence assumption, different uncertain factors are allowed to have different sample sizes and uncertainty sets. We provide two examples that can be described by our distributional uncertainty set as follows. For notational simplicity, we suppress the subscript k .

EXAMPLE 1 (MOMENT). Assume that $\mathbb{E}_{\mathbb{P}}[\tilde{z}^{m_q}] \in [\underline{\mu}_q, \bar{\mu}_q]$ for $q = 1, \dots, Q$, and m_q is a positive integer representing its moment. We consider the following instance of $\mathbf{g}$ and $\mathcal{C}$ in the distributional uncertainty set (4),

$$\mathbf{g}(z) = (z^{m_q})_{q=1, \dots, Q}, \quad G = [I, -I]', \quad H = \mathbf{0},$$

$$\mathbf{d} = [\bar{\mu}, -\underline{\mu}]', \quad \mathcal{H} = \mathbb{R}_+^{2Q}.$$

This allows us to characterize the random variables with any moment information.

EXAMPLE 2 (KULLBACK-LEIBLER DIVERGENCE 1951). Assume that $\sum_{l=1}^L p_l \ln(p_l/q_l) \leq \nu$ for a reference distribution $\mathbf{q} \in \mathbb{R}_+^L$, i.e., $\sum_{l=1}^L q_l = 1$, and an uncertainty level $\nu \in \mathbb{R}_+$, where $p_l = \mathbb{P}(\tilde{z} = z_l), l = 1, \dots, L$. This information can be represented by the uncertainty set $\mathcal{P}$. We let $\mathbf{g}(\tilde{z}) = (g_1(\tilde{z}), \dots, g_L(\tilde{z}))$ and for all $l = 1, \dots, L$,

$$g_l(\tilde{z}) = \begin{cases} 1 & \text{if } \tilde{z} = z_l, \\ 0 & \text{otherwise,} \end{cases}$$

and

$$\mathcal{C} = \left\{ (p_1, \dots, p_L): \exists \mathbf{h} \in \mathbb{R}^L \text{ such that } \sum_{l=1}^L h_l \leq \nu, \sum_{l=1}^L p_l = 1, p_l \exp(-h_l/p_l) \leq q_l, \forall l = 1, \dots, L \right\}.$$

Although the constraints $p_l \exp(-h_l/p_l) \leq q_l$ are not second-order cone representable, we can use second-order cone

constraints to approximate these constraints to any given level of computational precision. See Chen and Sim (2009).

3. URI

In this section, we specify an objective function for the project selection problem to evaluate the preference for project portfolio returns, $\tilde{\pi}(\mathbf{y})$, under our model of uncertainty.

To incorporate the effect of a target, which is an influential factor in decision making (Lanzillotti 1958; Mao 1970; Payne et al. 1980, 1981), we evaluate a project portfolio return by its target premium, i.e., the return minus target. Specifically, given a target return τ , we define $\mathcal{W} = \{\tilde{w}: \tilde{w} = \tilde{v} - \tau, \tilde{v} \in \mathcal{V}\}$ as the set of all possible target premiums. Hence the target premium $\tilde{w}$ with $\mathbb{P}(\tilde{w} \geq 0) = 1$ implies that the target return can be achieved almost surely. By optimizing over the risk of target premiums, we find a project portfolio whose uncertain return has lowest risk regarding the target τ .

Before introducing our approach to quantify the risk of target premium, we first review the concept of a *coherent risk measure*, which is widely used for evaluating risk (Artzner et al. 1999, Föllmer and Schied 2004). Let $\tilde{w} \geq \tilde{w}^o$ denote $\tilde{w}(\omega) \geq \tilde{w}^o(\omega)$ for all $\omega \in \Omega$, and $\tilde{w} \succ \tilde{w}^o$ represent that there exists an $\epsilon > 0$ such that $\tilde{w} \geq \tilde{w}^o + \epsilon$. A coherent risk measure is defined by a set of properties as follows.

DEFINITION 2 (ARTZNER ET AL. 1999). A function $\psi: \mathcal{W} \rightarrow \mathbb{R}$ is a *coherent risk measure* if for all $\tilde{w}, \tilde{w}^o \in \mathcal{W}$, it satisfies the following properties:

1. Monotonicity: if $\tilde{w} \geq \tilde{w}^o$, then $\psi(\tilde{w}) \leq \psi(\tilde{w}^o)$.
2. Cash Invariance: $\psi(\tilde{w} + \delta) = \psi(\tilde{w}) - \delta$ for all $\delta \in \mathbb{R}$.
3. Convexity: $\psi(\lambda \tilde{w} + (1 - \lambda)\tilde{w}^o) \leq \lambda \psi(\tilde{w}) + (1 - \lambda)\psi(\tilde{w}^o)$ for all $\lambda \in [0, 1]$.
4. Positive Homogeneity: $\psi(\lambda \tilde{w}) = \lambda \psi(\tilde{w})$ for all $\lambda > 0$.

Even though coherent risk measures have been widely applied (see, for example, Rockafellar and Uryasev 2000, Choi et al. 2011), they ignore the effect of the target. To demonstrate this, we consider any two uncertain returns, $\tilde{v}, \tilde{v}^o \in \mathcal{V}$. The difference between the coherent risk measure of their target premiums, $\tilde{v} - \tau$ and $\tilde{v}^o - \tau$, is

$$\begin{aligned} \psi(\tilde{v} - \tau) - \psi(\tilde{v}^o - \tau) &= (\psi(\tilde{v}) + \tau) - (\psi(\tilde{v}^o) + \tau) \\ &= \psi(\tilde{v}) - \psi(\tilde{v}^o), \end{aligned} \quad (5)$$

which is independent of the target τ . Therefore, if we are ranking $\tilde{v}$ and $\tilde{v}^o$ by the coherent risk measure of their target premiums, the target has no influence. From Equation (5), we observe that this independence of the target is due to the use of Cash Invariance, which enables the coherent risk measure to separate the target from the target premium. To resolve this issue, we replace the property of Cash Invariance by a target-sensitive property, and define the URI as follows.

DEFINITION 3. A function $\rho: \mathcal{W} \rightarrow [0, \infty]$ is a *URI* if for all $\tilde{w}, \tilde{w}^o \in \mathcal{W}$, it satisfies the following properties:

1. Monotonicity: if $\tilde{w} \geq \tilde{w}^o$, then $\rho(\tilde{w}) \leq \rho(\tilde{w}^o)$.
2. Satisficing:
 - (a) Attainment Content: if $\tilde{w} \geq 0$, then $\rho(\tilde{w}) = 0$;
 - (b) Starvation Aversion: if $\tilde{w} < 0$, then $\rho(\tilde{w}) = \infty$.
3. Convexity: $\rho(\lambda \tilde{w} + (1 - \lambda)\tilde{w}^o) \leq \lambda \rho(\tilde{w}) + (1 - \lambda)\rho(\tilde{w}^o)$ for all $\lambda \in [0, 1]$.
4. Positive Homogeneity: $\rho(\lambda \tilde{w}) = \lambda \rho(\tilde{w})$ for all $\lambda > 0$.

In Definition 3, the *Satisficing* property is specific to the target-based criterion. The first part, *Attainment Content*, implies that if an uncertain return can almost surely achieve the target, then it has zero URI. The second part, *Starvation Aversion*, implies that the URI is absolutely intolerant to any uncertain return where the target cannot be attained almost surely.

We now justify the new URI risk measure by relating it to previously discussed risk measures. First, we consider the classical convex risk measure.

THEOREM 1. A function $\rho: \mathcal{W} \rightarrow [0, \infty]$ is a *URI* if and only if it has the representation

$$\rho(\tilde{w}) = \inf \left\{ \frac{1}{\alpha} : \psi(\alpha \tilde{w}) \leq 0, \alpha > 0 \right\}, \quad (6)$$

where we define $\inf \emptyset = \infty$ by convention and ψ is a normalized convex risk measure, i.e., it satisfies the properties of Monotonicity, Cash Invariance, and Convexity, and has $\psi(0) = 0$. Conversely, given a *URI* ρ , the underlying normalized convex risk measure is given by

$$\psi(\tilde{w}) = \min \{a : \rho(\tilde{w} + a) \leq 1\}. \quad (7)$$

To understand the insight of the dual representation (6), we need the following result.

PROPOSITION 1. Given any target premium $\tilde{w} \in \mathcal{W}$ and a convex risk measure ψ normalized by $\psi(0) = 0$, there exists an $\alpha^* \in \mathbb{R}$ such that $\forall \alpha > 0$, $\psi(\alpha \tilde{w}) \leq 0$ if and only if $\alpha \leq \alpha^*$.

We observe that for any given target premium $\tilde{w}$ and a positive scale factor α , $\alpha \tilde{w}$ is acceptable, i.e., $\psi(\alpha \tilde{w}) \leq 0$, only when the scalar factor is no more than a benchmark value α^* . From Theorem 1, the URI can be interpreted as the reciprocal of this benchmark value. A small value of the URI is appealing since it corresponds to a high value of the benchmark α^* , which implies that the target premium remains acceptable for a large set of multiplicative perturbations.

Next, we establish a connection between the URI and the classical expected utility criterion. To do so, we consider the URI whose underlying convex risk measure is the *shortfall risk measure* (Föllmer and Schied 2002). Under distributional ambiguity, we redefine it as follows.

DEFINITION 4. A *shortfall risk measure* is any function $\psi_U: \mathcal{W} \rightarrow \mathbb{R}$ defined by

$$\psi_U(\tilde{w}) = \inf_{\eta \in \mathbb{R}} \left\{ \eta: \inf_{\mathbb{P} \in \mathcal{P}} \mathbb{E}_{\mathbb{P}}[u(\tilde{w} + \eta)] \geq 0 \right\},$$

where $u: \mathbb{R} \rightarrow \mathbb{R}$ is an increasing concave differentiable function normalized by $u(0) = 0$ and $u'(0) = 1$. Correspondingly, through the dual representation in Theorem 1, we define the utility-based URI, $\rho_U: \mathcal{W} \rightarrow [0, \infty]$, as one constructed based on a shortfall risk measure as follows:

$$\rho_U(\tilde{w}) = \inf \left\{ \frac{1}{\alpha}: \inf_{\mathbb{P} \in \mathcal{P}} \mathbb{E}_{\mathbb{P}}[u(\alpha \tilde{w})] \geq 0, \alpha > 0 \right\}.$$

We can interpret the shortfall risk measure, $\psi_U(\tilde{w})$, as the minimum cash amount, η , that needs to be added to the target premium $\tilde{w}$ such that the overall expected utility exceeds zero. This can be shown to be a convex risk measure. Correspondingly, the utility-based URI is the reciprocal of the largest positive scalar factor α such that the expected utility of $\alpha \tilde{w}$ remains positive.

REMARK 2. Given known distributions, when we use exponential utility, i.e., $u(x) = 1 - \exp(-x)$, the URI coincides with the *riskiness index* of Aumann and Serrano (2008); when $\tilde{w} \geq 0 \forall \tilde{w} \in \mathcal{W}$ and we use log utility, i.e., $u(x) = \log(1 + x)$, the corresponding URI is the *operational measure of riskiness* of Foster and Hart (2009). Finally, the URI falls within the framework of *coherent satisfying measures* of Brown and Sim (2009).

The project selection problem under the utility-based URI can be written as

$$\rho^* = \min_{\mathbf{y} \in \mathcal{Y}} \rho_U(\tilde{\pi}(\mathbf{y}) - \tau).$$

As we discuss in §1, the only utility function that can be tractably evaluated for the summation of independently distributed random variables is the exponential function. This exponential utility function is of special interest since it is associated with a constant ARA, the value of which is α and insensitive to the payoff x . This specific form of utility allows us to develop a tractable model and provide a computationally efficient solution procedure. Hence we define the EURI as follows.

DEFINITION 5. Given a target premium $\tilde{w} \in \mathcal{W}$, the EURI $\rho_E: \mathcal{W} \rightarrow [0, \infty]$ is defined by

$$\begin{aligned} \rho_E(\tilde{w}) &= \inf \left\{ \frac{1}{\alpha}: \frac{1}{\alpha} \inf_{\mathbb{P} \in \mathcal{P}} \mathbb{E}_{\mathbb{P}}(1 - \exp(-\alpha \tilde{w})) \geq 0, \alpha > 0 \right\} \\ &= \inf \left\{ \frac{1}{\alpha}: C_{\alpha}(\tilde{w}) \geq 0, \alpha > 0 \right\}, \end{aligned}$$

where

$$C_{\alpha}(\tilde{w}) = -\frac{1}{\alpha} \ln \left(\sup_{\mathbb{P} \in \mathcal{P}} \mathbb{E}_{\mathbb{P}}(\exp(-\alpha \tilde{w})) \right).$$

Function $C_{\alpha}(\tilde{w})$ is the certainty equivalent of the uncertain target premium $\tilde{w}$ under the worst-case exponential utility of Gilboa and Schmeidler (1989) with ARA parameter α . Hence, the EURI is the reciprocal of the highest ARA for which the certainty equivalent of target premium $\tilde{w}$ under ambiguity aversion equals 0. This index extends the riskiness index of Aumann and Serrano (2008) to incorporate the target and distributional ambiguity. We now present some useful properties of function $C_{\alpha}(\cdot)$.

REMARK 3. For any $\tilde{w} \in \mathcal{W}$, $C_{\alpha}(\tilde{w})$ is nonincreasing in α . Moreover, $C_{\alpha}(\tilde{w} + \tilde{w}^o) = C_{\alpha}(\tilde{w}) + C_{\alpha}(\tilde{w}^o)$ if $\tilde{w}$ and $\tilde{w}^o$ are independent from each other.

Given this exponential utility function, besides the properties introduced in Definition 3, $\rho_E(\tilde{w})$ also provides an envelope for the UP.

PROPOSITION 2. Given any target premium $\tilde{w} \in \mathcal{W}$, the value of $\rho^* = \rho_E(\tilde{w})$ defines the following envelope for the UP:

$$\mathbb{P}(\tilde{w} \leq -\theta) \leq \exp(-\theta/\rho^*), \quad \forall \theta \geq 0.$$

This proposition explains how the EURI relates to the UP. The EURI controls the UP for any target level. This also shows that, unlike the UP approach, the EURI is sensitive to the extent of violation of the target level.

From now on, we focus on the project selection problem under the EURI

$$\rho^* = \min_{\mathbf{y} \in \mathcal{Y}} \rho_E(\tilde{\pi}(\mathbf{y}) - \tau).$$

Since the target τ is exogenous in our model, setting its value is an important issue in practice, but one that has not been widely studied. Simon (1955) provides an example of selling a house, where the agent's target level is determined from learning about the housing market. Similarly, the project selection committee can determine a range for the target from an understanding of the investment climate specific to the industry. For further specification within this range, a field study by Merchant and Manzoni (1989) shows that targets are typically set at a level that can be achieved with 80%–90% probability. Normative research focuses mainly on the motivational effects of target setting on employee performance (Locke and Latham 2002). Since this issue is beyond the scope of our work, we assume that the target τ is given. However, to ensure that the model is meaningful, we assume that the target is reasonably set, such that the target is neither unachievable in expectation nor achievable almost surely. We formalize these assumptions as follows.

ASSUMPTION 1. There exists a feasible project selection $\mathbf{y} \in \mathcal{Y}$ such that the expected return of the projects under ambiguity aversion exceeds the target, i.e.,

$$\inf_{\mathbb{P} \in \mathcal{P}} \mathbb{E}_{\mathbb{P}}(\tilde{\pi}(\mathbf{y})) > \tau.$$

ASSUMPTION 2. *There does not exist a feasible project selection $\mathbf{y} \in \mathcal{Y}$ such that the total uncertain return of the selected projects can achieve the target almost surely, i.e., $\tilde{\pi}(\mathbf{y}) \geq \tau$.*

From Assumption 1, we can guarantee that there exists a feasible solution $\mathbf{y} \in \mathcal{Y}$ such that $\rho_E(\tilde{\pi}(\mathbf{y}) - \tau) < \infty$, hence $\rho^* < \infty$. From Assumption 2, since no solution can achieve the target almost surely, we have $\rho^* > 0$.

We exclude project portfolio selections whose URI is infinite, and define

$$\hat{\mathcal{Y}} = \left\{ \mathbf{y} \in \mathcal{Y} : \inf_{\mathbb{P} \in \mathcal{P}} \mathbb{E}_{\mathbb{P}}(\tilde{\pi}(\mathbf{y})) > \tau \right\}.$$

It is easy to show that $\hat{\mathcal{Y}}$ is nonempty, and for all $\mathbf{y} \in \hat{\mathcal{Y}}$, $\rho_E(\tilde{\pi}(\mathbf{y}) - \tau) \in (0, \infty)$,

$$\rho^* = \min_{\mathbf{y} \in \hat{\mathcal{Y}}} \rho_E(\tilde{\pi}(\mathbf{y}) - \tau). \quad (8)$$

4. Solution Procedure

In this section, we describe a procedure to solve Problem (8). Our first result linearizes the indicator function, $\Gamma(\beta, \mathbf{y})$.

LEMMA 1.

$$\sum_{\beta \in \mathcal{C}} r_{\beta}^k \Gamma(\beta, \mathbf{y}) = \mathbf{x}' \mathbf{v}^k, \quad k = 0, \dots, K,$$

where $\mathbf{x}$ is the unique binary solution to the equation system

$$x_{\beta} \leq y_i, \quad \beta \in \mathcal{C}, i \in \beta$$

$$x_{\beta} + |\beta| \geq \sum_{i \in \beta} y_i + 1, \quad \beta \in \mathcal{C}$$

$$\mathbf{x} \in \{0, 1\}^{|\mathcal{C}|};$$

$$\mathbf{v}^k = (v_{\beta}^k)_{\beta \in \mathcal{C}}, \quad \text{and} \quad v_{\beta}^k = r_{\beta}^k - \sum_{\beta' \subset \beta, \beta' \in \mathcal{C}} v_{\beta'}^k, \quad k = 0, \dots, K.$$

Lemma 1 implies that the total return can be written as a linear function of uncertain factors.

$$\tilde{\pi}(\mathbf{y}) = \sum_{k=0}^K \mathbf{x}' \mathbf{v}^k \tilde{z}_k, \quad (9)$$

where $\mathbf{x}$ is determined from $\mathbf{y}$ according to Lemma 1, and we let $\tilde{z}_0 = 1$ for simplicity. Hence, Problem (8) can be reformulated as

$$\rho^* = \min_{\mathbf{x} \in \mathcal{X}} U_E(\mathbf{x}), \quad (10)$$

where we define

$$\mathcal{X} = \left\{ \mathbf{x} : \begin{array}{l} x_{\beta} \leq y_i, \quad \beta \in \mathcal{C}, i \in \beta \\ x_{\beta} + |\beta| \geq \sum_{i \in \beta} y_i + 1, \quad \beta \in \mathcal{C} \\ \mathbf{y} \in \hat{\mathcal{Y}}, \\ \inf_{\mathbb{P} \in \mathcal{P}} \mathbb{E}_{\mathbb{P}} \left(\sum_{k=0}^K \mathbf{x}' \mathbf{v}^k \tilde{z}_k \right) > \tau, \\ \mathbf{x} \in \{0, 1\}^{|\mathcal{C}|}, \end{array} \right\}$$

and

$$\begin{aligned} U_E(\mathbf{x}) &= \inf \left\{ \gamma > 0 : C_{1/\gamma} \left(\sum_{k=0}^K \mathbf{x}' \mathbf{v}^k \tilde{z}_k - \tau \right) \geq 0 \right\} \\ &= \inf \left\{ \gamma > 0 : \sum_{k=0}^K C_{1/\gamma}(\mathbf{x}' \mathbf{v}^k \tilde{z}_k) \geq \tau \right\}. \end{aligned}$$

The last equality follows from Remark 3. Note that in the definition of $\mathcal{X}$, since the feasible $\mathbf{x}$ is integer, there must exist a small positive scalar ϵ such that we can replace the constraint $\inf_{\mathbb{P} \in \mathcal{P}} \mathbb{E}_{\mathbb{P}}(\sum_{k=0}^K \mathbf{x}' \mathbf{v}^k \tilde{z}_k) > \tau$ by $\inf_{\mathbb{P} \in \mathcal{P}} \mathbb{E}_{\mathbb{P}}(\sum_{k=0}^K \mathbf{x}' \mathbf{v}^k \tilde{z}_k) \geq \tau + \epsilon$, but $\mathcal{X}$ remains the same.

Since $\mathbf{x}$ is a vector of binary decision variables, and $U_E(\mathbf{x})$ is nonlinear in $\mathbf{x}$, we cannot solve the problem directly. Next, we transform $U_E(\mathbf{x})$ into a piecewise linear function, and use Benders decomposition to solve Problem (10).

PROPOSITION 3. *For all $\mathbf{x} \in \mathcal{X}$,*

$$U_E(\mathbf{x}) = \max_{\boldsymbol{\theta} \in \mathcal{Z}} \{U_E(\boldsymbol{\theta}) + \mathbf{D}_E(\boldsymbol{\theta})'(\mathbf{x} - \boldsymbol{\theta})\},$$

where $\mathbf{D}_E(\boldsymbol{\theta})$ is a subgradient defined by

$$\begin{aligned} \mathbf{D}_E(\boldsymbol{\theta}) &= \sum_{k=0}^K \frac{\mathbb{E}_{\mathbb{P}_k}(\exp(-\boldsymbol{\theta}' \mathbf{v}^k \tilde{z}_k / \gamma_{\theta}))}{\mathbb{E}_{\mathbb{P}_k}(\exp(-\boldsymbol{\theta}' \mathbf{v}^k \tilde{z}_k / \gamma_{\theta}))} \mathbf{v}^k \\ &\quad \cdot \left[\sum_{k=0}^K \left(\ln \mathbb{E}_{\mathbb{P}_k} \left(\exp \left(-\frac{\boldsymbol{\theta}' \mathbf{v}^k \tilde{z}_k}{\gamma_{\theta}} \right) \right) \right. \right. \\ &\quad \left. \left. + \frac{\mathbb{E}_{\mathbb{P}_k}(\exp(-\boldsymbol{\theta}' \mathbf{v}^k \tilde{z}_k / \gamma_{\theta})) \boldsymbol{\theta}' \mathbf{v}^k \tilde{z}_k}{\gamma_{\theta} \mathbb{E}_{\mathbb{P}_k}(\exp(-\boldsymbol{\theta}' \mathbf{v}^k \tilde{z}_k / \gamma_{\theta}))} \right) \right]^{-1}, \end{aligned}$$

$\gamma_{\theta} = U_E(\boldsymbol{\theta})$, $\mathbb{P}_k$ is a probability distribution with

$$\mathbb{P}_k \in \arg \sup_{\mathbb{P} \in \mathcal{P}_k} \mathbb{E}_{\mathbb{P}}(\exp(-\boldsymbol{\theta}' \mathbf{v}^k \tilde{z}_k / \gamma)), \quad k = 1, \dots, K,$$

and $\mathbb{P}_0(\tilde{z}_0 = 1) = 1$.

Observe that the transformation of $U_E(\mathbf{x})$ into a piecewise linear function in Proposition 3 uses all the points $\boldsymbol{\theta} \in \mathcal{Z}$, and hence is exact. We evaluate $U_E(\boldsymbol{\theta})$ and $\mathbf{D}_E(\boldsymbol{\theta})$ by computing the worst-case distribution $\mathbb{P}_k$, $k = 1, \dots, K$ with the following result.

PROPOSITION 4. *Given any $\boldsymbol{\theta} \in \mathcal{Z}$, $\gamma \in (0, \infty)$, and $k = 1, \dots, K$, the distribution*

$$\mathbb{P}_k \in \arg \inf_{\mathbb{P} \in \mathcal{P}_k} \mathbb{E}_{\mathbb{P}}(\exp(-\boldsymbol{\theta}' \mathbf{v}^k \tilde{z}_k / \gamma))$$

can be characterized by $\mathbb{P}_k(\tilde{z}_k = z_{kl}) = p_{kl}$, $l = 1, \dots, L_k$, where $p_{k1}, \dots, p_{kL_k}$ solves the optimization problem

$$\begin{aligned} \sup \quad & \sum_{l=1}^{L_k} p_{kl} \exp \left(-\frac{\boldsymbol{\theta}' \mathbf{v}^k z_{kl}}{\gamma} \right) \\ \text{s.t.} \quad & \sum_{l=1}^{L_k} p_{kl} G_k \mathbf{g}_k(z_{kl}) + H_k \mathbf{h} \preceq_{\mathcal{H}_k} \mathbf{d}_k, \\ & \sum_{l=1}^{L_k} p_{kl} = 1, \\ & p_{kl} \geq 0, \quad \forall l = 1, \dots, L_k. \end{aligned}$$

This is a conic optimization problem that can be solved via commercial solvers. Using Proposition 4, we evaluate the worst-case distribution, and find $\gamma_\theta = U_E(\theta)$ using a binary search method as justified by Proposition 1. Based on the value of γ_θ , we use Proposition 4 again to find $D_E(\theta)$.

Now, Problem (10) can be formulated as the mixed-integer program

$$\begin{aligned} \min_{\mathbf{x}, t} \quad & t \\ \text{s.t.} \quad & t \geq U_E(\theta) + D_E(\theta)'(\mathbf{x} - \theta), \quad \forall \theta \in \mathcal{X}, \\ & \mathbf{x} \in \mathcal{X}. \end{aligned} \quad (11)$$

Nevertheless, the size of $\mathcal{X}$ is large since it contains all feasible selections, hence the existence of an efficient algorithm for searching it is unlikely. This justifies the following enumerative procedure.

Algorithm BD

1. Start with any nonempty subset $\mathcal{X}^o \subset \mathcal{X}$.
2. Solve the following subproblem:

$$\begin{aligned} \min_{\mathbf{x}, t} \quad & t \\ \text{s.t.} \quad & t \geq U_E(\theta) + D_E(\theta)'(\mathbf{x} - \theta), \quad \forall \theta \in \mathcal{X}^o \\ & \mathbf{x} \in \mathcal{X} \end{aligned} \quad (12)$$

and denote the solution by $\mathbf{x}^*$ and t^* .

3. If $t^* = U_E(\mathbf{x}^*)$, then output the optimal value and optimal solution $\mathbf{x}^*$, and stop; otherwise, add $\mathbf{x}^*$ into $\mathcal{X}^o$, and go to Step 2.

PROPOSITION 5. *Algorithm BD finds an optimal solution to Problem (11) in a finite number of steps.*

Under the EURI, due to the separability of independent uncertainties described in Remark 3, we calculate the worst-case distribution for different uncertain factors independently, and hence significantly reduce the dimension of the optimization problem.

5. Solvability, and Heuristic for a Special Case

In this section, we investigate the computational solvability of several versions of the problem and provide a heuristic for solving the special case with only a single budget constraint. Given the definition of the EURI and the independence property of the worst-case certainty equivalent from Remark 3, the project selection problem can be formulated as follows:

$$\begin{aligned} \inf \quad & \frac{1}{\alpha} \\ \text{s.t.} \quad & \sum_{k=0}^K C_\alpha(\mathbf{x}' \mathbf{v}^k \tilde{z}_k) \geq \tau, \\ & \mathbf{x} \in \mathcal{X}, \\ & \alpha > 0. \end{aligned}$$

Observe that function $C_\alpha(\cdot)$ is monotonic in α , hence we can perform a binary search for α . For any given $\alpha > 0$, we only need to solve the subproblem as follows:

$$\max_{\mathbf{x} \in \mathcal{X}} \sum_{k=0}^K C_\alpha(\mathbf{x}' \mathbf{v}^k \tilde{z}_k). \quad (13)$$

5.1. Solvability

In this section, we consider the subproblem (13) and discuss its computational complexity.

REMARK 4. In some cases, the structure of the constraint matrix $\mathbf{A}\mathbf{y} \leq \mathbf{b}$ implies intractability, even for a deterministic version of the project selection problem. For example, under a simple knapsack constraint, the recognition version of the deterministic problem is binary *NP*-complete (Garey and Johnson 1979). Further, under arbitrary resource constraints, the recognition version of the deterministic problem is unary *NP*-complete.

We therefore focus on solvability results that derive from the uncertainty and interaction structure of the problem.

Independent Returns Without Interactions

PROPOSITION 6. *If there are no interactions among the projects, and their returns are independent, then the objective function in Problem (13) is a linear function of $\mathbf{y}$.*

Proposition 6 shows that, without interactions and with independent uncertain returns, Problem (13) retains the solvability of the problem $\max_{\mathbf{y} \in \mathcal{Y}} \mathbf{r}'\mathbf{y}$. For example, if the feasible set of $\mathbf{y}$ is a uniform matroid, i.e., $\mathcal{Y} = \{\mathbf{y} \in \{0, 1\}^N: \sum_{k \in \mathcal{N}} y_k = m\}$, then Problem (13) is optimally solved in $O(N \log N)$ time by sorting $(r_i^0 + \sum_{k \in \mathcal{F}_i} C_\alpha(\tilde{z}_k r_i^k))$, $i \in \mathcal{N}$ in nonincreasing order and selecting the first m projects. Here, $\mathcal{F}_i = \{k: r_i^k \neq 0, k = 1, \dots, K\}$, $i \in \mathcal{N}$. Alternatively, if the feasible set of $\mathbf{y}$ is defined by a simple knapsack constraint, i.e., $\mathcal{Y} = \{\mathbf{y} \in \{0, 1\}^N: \sum_{k \in \mathcal{N}} a_k y_k = b\}$, then Problem (13) is optimally solved in pseudopolynomial time (Kellerer et al. 2004).

COROLLARY 1. *For the special case where project returns are independent and no interactions exist, Problem (13) with a single budget constraint can be simplified to*

$$\begin{aligned} \max \quad & \sum_{i \in \mathcal{N}} \left(r_i^0 + \sum_{k \in \mathcal{F}_i} C_\alpha(r_i^k \tilde{z}_k) \right) y_i \\ \text{s.t.} \quad & \mathbf{c}'\mathbf{y} \leq b, \\ & \mathbf{y} \in \{0, 1\}^N. \end{aligned}$$

Correlated Returns Without Interactions

We assume the absence of interactions, to focus solely on the role of correlation. Hence $\mathcal{C} = \mathcal{N}$. We have the following result.

PROPOSITION 7. *If there are no interactions among the projects, then the recognition version of Problem (13) with correlated returns under a uniform matroid is binary *NP*-complete.*

Independent Returns and Interactions

PROPOSITION 8. *If there are interactions among the projects, then the recognition version of Problem (13) with deterministic returns under a uniform matroid is unary NP-complete.*

5.2. Heuristic EURI for a Special Case

In many situations, project selection is mainly constrained by a budget. In that case, the feasible project selection set is

$$\mathcal{Y} = \left\{ \mathbf{y}: \begin{array}{l} \mathbf{c}'\mathbf{y} \leq b \\ \mathbf{y} \in \{0, 1\}^N \end{array} \right\}.$$

Let $\mathbf{e}_i$ represent an N dimensional vector, where the i th element is one, and the others are zero. We describe a simple heuristic, which can be implemented on a spreadsheet for Problem (13).

Heuristic EURI

Input: ARA parameter α .

Output: Heuristic solution $\mathbf{y}_\alpha^G$, and its certainty equivalent C_α^G .

1. Start with an empty selection, $\mathbf{y} = \mathbf{0}$;

Set $\bar{c} = 0$ to represent the total cost of the currently selected projects.

2. Let $\mathcal{J} = \{i: i \in \mathcal{N}, y_i = 0, \bar{c} + c_i \leq b\}$. If $\mathcal{J} = \emptyset$, then go to Step 5.

3. Find

$$j \in \arg \max \left\{ \frac{C_\alpha(\tilde{\pi}(\mathbf{y} + \mathbf{e}_i)) - C_\alpha(\tilde{\pi}(\mathbf{y}))}{c_i}, i \in \mathcal{J} \right\}.$$

4. If $C_\alpha(\tilde{\pi}(\mathbf{y} + \mathbf{e}_j)) > C_\alpha(\tilde{\pi}(\mathbf{y}))$, then select the j th project, set $\mathbf{y} = \mathbf{y} + \mathbf{e}_j$ and $\bar{c} = \bar{c} + c_j$, and go to Step 2; otherwise, go to Step 5.

5. Output $\mathbf{y}_\alpha^G = \mathbf{y}$ and $C_\alpha^G = C_\alpha(\tilde{\pi}(\mathbf{y}))$, and stop.

This heuristic is motivated by the greedy heuristic for the classical zero-one knapsack problem (Kellerer et al. 2004). At each stage, we select the project that returns the largest ratio of increase in certainty equivalent to project cost. In the appendix, we provide a simple example to illustrate the calculation procedure. Computational results for Heuristic EURI appear in §6.

6. Computational Studies

In §6.1, we describe several benchmark approaches from the project selection literature, and how to solve them. Section 6.2 contains a comparative study of the EURI and Heuristic EURI against the benchmark approaches. Section 6.3 provides a sensitivity analysis of the performance of the EURI. Finally, §6.4 studies the robustness of the EURI. All models are solved using CPLEX 11.0 on a computer with Intel I5 processor and 8 GB RAM.

6.1. Benchmark Selection Approaches

In this set of experiments, we assume for the EURI approaches and the benchmarks that the probability distributions of the factors are known. In §6.4, we show that this assumption does not significantly affect the performance of the EURI approaches. We now describe several benchmark selection approaches for comparison with the EURI.

6.1.1. Expected Return. We consider maximizing expected return as the objective. Based on the formulation of total return in Equation (9), we need to solve the following problem:

$$\max_{\mathbf{x} \in \mathcal{X}} \mathbb{E} \left(\sum_{k=0}^K \mathbf{x}' \mathbf{v}^k \tilde{z}_k \right),$$

which is equivalent to

$$\max_{\mathbf{x} \in \mathcal{X}} \sum_{k=0}^K \mathbf{x}' \mathbf{v}^k \mathbb{E}(\tilde{z}_k). \quad (14)$$

Problem (14) is a deterministic problem, which is computationally easy to solve for large instances.

6.1.2. Underperformance Probability. We consider the objective of minimizing the probability that the return will fall below a given target, τ . This model is

$$\min_{\mathbf{x} \in \mathcal{X}} \mathbb{P} \left(\sum_{k=0}^K \mathbf{x}' \mathbf{v}^k \tilde{z}_k < \tau \right).$$

However, optimization of this model is a highly intractable problem. We solve it using simulation. We generate a sample consisting of M independent random factor scenarios. Each scenario i contains a realization of independently generated random factors $z_1^i, \dots, z_K^i$. Given $\mathbf{v}^0, \dots, \mathbf{v}^K$, the problem of minimizing the UP is

$$\begin{aligned} \min \quad & \frac{1}{M} \sum_{i=1}^M I_i \\ \text{s.t.} \quad & I_i \geq \frac{\tau - \sum_{k=0}^K \mathbf{x}' \mathbf{v}^k z_k^i}{G}, \quad i = 1, \dots, M \\ & I_i \in \{0, 1\}, \quad i = 1, \dots, M \\ & \mathbf{x} \in \mathcal{X}, \end{aligned} \quad (15)$$

where G is a sufficiently large scalar that

$$\frac{\tau - \sum_{k=0}^K \mathbf{x}' \mathbf{v}^k z_k^i}{G} \in (-1, 1).$$

If the sample size M is large enough, then the solution from Problem (15) closely approximates the solution that minimizes the UP.

6.1.3. Markowitz Model. We use the Markowitz (1959) model, which yields portfolio returns with minimum variance, subject to the expected return being greater than a target. This problem can be formulated as

$$\begin{aligned} \min \quad & \text{Var} \left(\sum_{k=0}^K \mathbf{x}' \mathbf{v}^k \tilde{z}_k \right) \\ \text{s.t.} \quad & \mathbb{E} \left(\sum_{k=0}^K \mathbf{x}' \mathbf{v}^k \tilde{z}_k \right) \geq \tau, \\ & \mathbf{x} \in \mathcal{X}, \end{aligned} \quad (16)$$

where $\text{Var}(\cdot)$ is the variance of a random variable.

Let Σ denote the covariance matrix of the random factors $\tilde{z}_0, \dots, \tilde{z}_K$. By assumption $\tilde{z}_k$ is independent of $\tilde{z}_i, i \neq k$, hence Σ is a diagonal matrix with k th diagonal element $\Sigma_{k,k} = \text{Var}(\tilde{z}_k)$. Therefore, Problem (16) is equivalent to

$$\begin{aligned} \min \quad & \mathbf{x}' \mathbf{V} \Sigma \mathbf{V}' \mathbf{x} \\ \text{s.t.} \quad & \sum_{k=0}^K \mathbf{x}' \mathbf{v}^k \mathbb{E}(\tilde{z}_k) \geq \tau, \\ & \mathbf{x} \in \mathcal{X}. \end{aligned} \quad (17)$$

Problem (17) is a quadratic optimization problem with binary decision variables, which can be solved using CPLEX.

6.1.4. Maximization of Roy's Safety-First Ratio. We consider maximization of Roy's safety-first ratio (1952) as the project selection criterion, and formulate the problem as

$$\max_{\mathbf{x} \in \mathcal{X}} \frac{\mathbb{E}\left(\sum_{k=0}^K \mathbf{x}' \mathbf{v}^k \tilde{z}_k\right) - \tau}{\sigma\left(\sum_{k=0}^K \mathbf{x}' \mathbf{v}^k \tilde{z}_k\right)}, \quad (18)$$

where $\sigma(\cdot)$ is the standard deviation of a random variable. To ensure that Roy's safety-first ratio optimization problem is well posed, we assume that the optimum Roy's safety-first ratio is positive and finite. Moreover, there does not exist a solution, $\mathbf{x} \in \mathcal{X}$, such that $\mathbb{E}(\sum_{k=0}^K \mathbf{x}' \mathbf{v}^k \tilde{z}_k) \geq \tau$ and $\sigma(\sum_{k=0}^K \mathbf{x}' \mathbf{v}^k \tilde{z}_k) = 0$. Next, we propose a solution procedure to solve this nonconvex discrete optimization problem. We define

$$Z(s) = \max_{\mathbf{x} \in \mathcal{X}} \mathbb{E}\left(\sum_{k=0}^K \mathbf{x}' \mathbf{v}^k \tilde{z}_k\right) - \tau - s \sigma\left(\sum_{k=0}^K \mathbf{x}' \mathbf{v}^k \tilde{z}_k\right). \quad (19)$$

PROPOSITION 9. *The function, $Z(s)$ is a continuous, nonincreasing function of $s > 0$. If Problem (18) is well posed, then*

- (i) *there exists an $s > 0$ such that $Z(s) = 0$, and*
- (ii) *the optimal solution of Problem (19) in which $Z(s) = 0$ is also optimal in Problem (18).*

As the result of Proposition 9, we find the optimum solution of Roy's safety-first ratio problem by performing a binary search on s until $Z(s) = 0$. Subproblem (19) is a second-order conic optimization problem with binary decision variables, which we solve to optimality using CPLEX.

6.2. Comparison with Benchmarks

Using the guidelines of Hall and Posner (2001), (a) we generate a wide range of parameter specifications, (b) the data generated is representative of real-world scenarios, and (c) the experimental design varies only in the parameters that may affect the analysis. A *project instance* is specified by a set of projects, deterministic constraints, correlations, random factors, and interactions. We randomly generate

200 project instances with $N = 50$, and a single budget constraint, which implies that no more than 25 projects can be selected.

We generate interactions by randomly assigning an index from $UI(1, \dots, 100)$ to each project. We specify that interaction effects $\tilde{r}_\beta = \eta_{|\beta|} \sum_{i \in \beta} \tilde{r}_i$ exist among projects with the same index. We let $\eta_1 = 1$, $\eta_2 = 1.1$, $\eta_3 = 1.25$, and $\eta_{|\beta|} = 1.5$ when $|\beta| \geq 4$. We consider 50 independent random factors, which follow two-point distributions with mean μ_i and standard deviation σ_i , given by $\mu_i = 80 + (i-1) \times 2.5$ and $\sigma_i = 10 + (i-1) \times 7.5$, $i = 1, \dots, 50$. To consider skewness, the probability of a high return for random factor i is independently generated by $p_i^H \sim U(0.55, 0.95)$. Given μ_i , σ_i , and p_i^H , the random factors are fully characterized. The total return of project i is $\tilde{r}_i = \sum_{k=0}^K r_i^k \tilde{z}_k$. We let $r_i^i = 1$, which specifies that project i depends mainly on its local factor i , and generate r_i^k as $U(-0.5, 0.5)$ for $k \neq i$, which specifies the correlation. Finally, let φ denote the ratio of the target to the maximum expected return, where $\varphi \in \{0.6, 0.7, 0.8\}$.

For each project instance, we (a) find solutions using each selection approach, (b) randomly generate a sample of size 50,000 for the 50 random factors, and (c) compute the returns for each selection approach and each sample instance.

To provide a fair comparison between our two models and the four benchmark approaches, we first consider the four criteria that the benchmark approaches directly optimize. These criteria are expected return, UP, standard deviation, and Roy's safety-first ratio, respectively. We also consider four widely used performance criteria that are optimized neither by the benchmark approaches nor by the models we propose. The first is expected loss (EL) relative to the target. Second, normalizing by the probability of a loss, we consider the conditional expected loss (CEL). Both these criteria are widely used in financial risk management (Embrechts et al. 1997). Finally, we consider the value-at-risk (VaR), i.e., the threshold loss that the project portfolio does not exceed with a specified probability, at the 95% and 99% levels (Jorion 2006). A negative VaR represents the minimum profit that is attainable with the specified probability. Table 1 shows the mean performance of each project selection approach for all eight criteria. Because of the slow convergence of Problem (15), the program for UP is terminated after 10 minutes for each instance. Hence, Problem (15) may not be solved to optimality. The average computation time to solve one instance under the EURI approach is about 8 seconds, compared to 0.04 seconds under the expected return maximization approach. However, the EURI time is still acceptable in practice, and much shorter than the Markowitz approach (350 seconds) or Roy's safety-first ratio approach (780 seconds).

Table 1 shows that the average performance of the EURI and Heuristic EURI is almost identical across all criteria. Hence, for the single budget constraint case, almost the

Table 1. Performance of various project selection approaches.

φ	Selection approach	Criterion							
		Expected return	UP (%)	Standard deviation	Roy's SF ratio	EL	CEL	VaR @ 95%	VaR @ 99%
0.6	EURI	12,018	18.67	3,830	0.8969	450	2,351	−5,400	−2,380
	Heuristic EURI	11,961	19.01	3,835	0.8818	463	2,373	−5,326	−2,294
	Expected return	14,367	17.50	6,275	0.9438	716	3,906	−3,428	1,623
	UP	13,271	16.70	4,849	0.9785	521	3,003	−4,798	−846
	Markowitz model	8,647	47.40	2,392	0.0123	943	1,990	−4,500	−2,583
	Roy's SF ratio	13,737	15.40	5,040	1.0371	489	3,055	−4,948	−868
0.7	EURI	13,126	24.21	4,546	0.6836	738	3,002	−5,249	−1,647
	Heuristic EURI	13,117	24.36	4,548	0.6787	743	3,008	−5,236	−1,628
	Expected return	14,367	23.39	6,275	0.7079	1,007	4,183	−3,428	1,623
	UP	13,646	23.60	5,239	0.6974	849	3,527	−4,470	−169
	Markowitz model	10,082	47.43	2,968	0.0091	1,174	2,476	−4,923	−2,536
	Roy's SF ratio	14,061	22.09	5,454	0.7537	817	3,593	−4,539	−117
0.8	EURI	13,884	30.84	5,243	0.4643	1,166	3,743	−4,764	−570
	Heuristic EURI	13,865	31.00	5,259	0.4593	1,179	3,764	−4,712	−497
	Expected return	14,367	30.49	6,275	0.4719	1,393	4,494	−3,428	1,623
	UP	13,682	32.73	5,467	0.4031	1,335	4,022	−4,115	360
	Markowitz model	11,521	47.41	3,683	0.0080	1,459	3,077	−5,113	−2,133
	Roy's SF ratio	14,242	29.89	5,768	0.4895	1,248	4,113	−4,173	492

Note. UP = underperformance probability; SF = safety first; EL = expected loss; CEL = conditional expected loss; VaR = value-at-risk.

full benefit of our models can be achieved by a simple spreadsheet-based solution approach.

We compare the EURI against the four benchmark approaches using the mean performance for the three target levels. Comparison with the risk-neutral maximization of expected return approach shows that the EURI is risk averse, evidenced by a 27.6% smaller standard deviation, a 24.4% smaller EL, and a 49.9% smaller VaR @ 95%. Comparison with the minimization of UP shows that the EURI is 12.3% smaller in standard deviation, 16.0% smaller in EL, and 15.5% smaller in VaR @ 95%. Comparison with the highly risk-averse Markowitz model shows that the EURI has a 48.2% smaller UP, a 70 times greater Roy's safety-first ratio, and a 34.2% smaller EL. Comparison with Roy's safety-first ratio maximization shows that the EURI is 16.2% smaller in standard deviation, 15.5% smaller in CEL, and eight times smaller in VaR @ 99%. In general, the EURI provides solutions with high Roy's safety-first ratio, low EL and CEL, and low VaR.

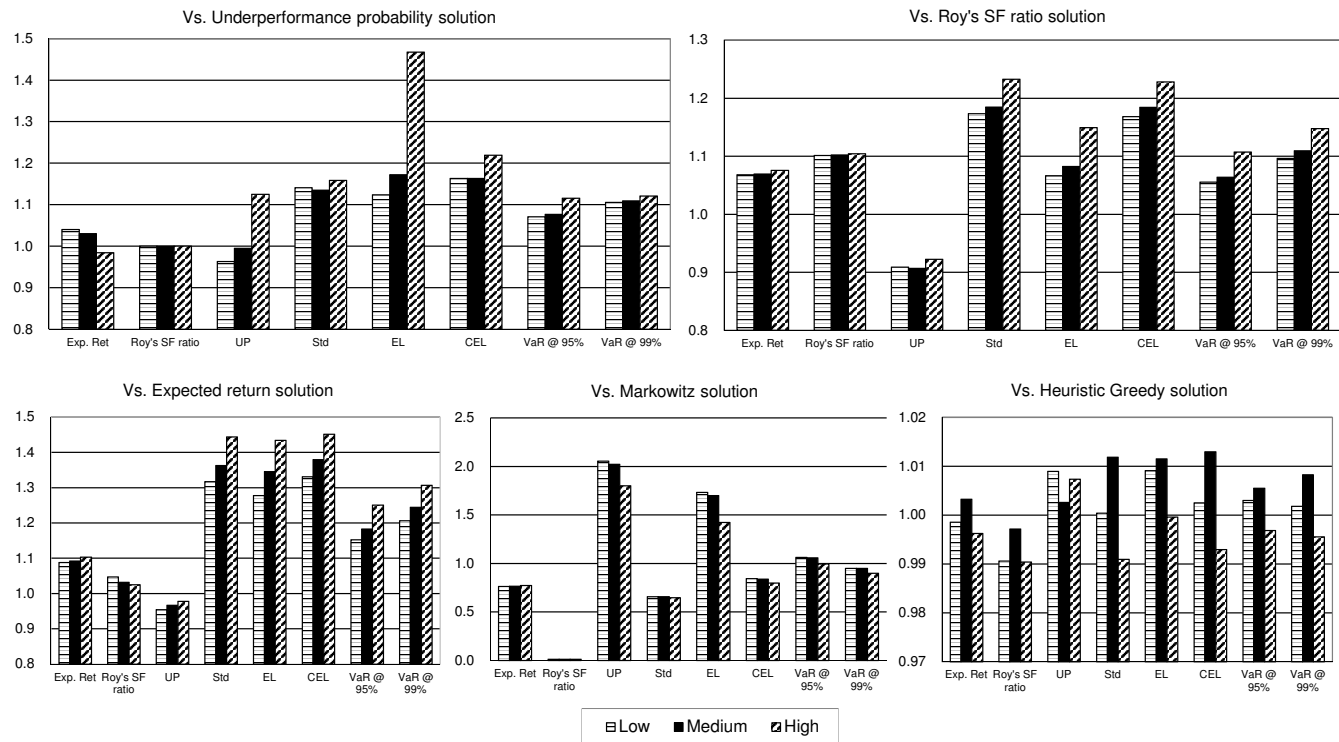
The only comparisons where our models fall substantially short are in expected return against the risk-neutral maximization of expected return approach, and in minimization of standard deviation against the highly risk-averse Markowitz model. In both cases, this occurs because the EURI and Heuristic EURI are mildly risk averse. The EURI and Heuristic EURI provide much better EL and VaR performance than the maximization of expected return, and much better expected return and UP than the Markowitz model. The expected return and Roy's safety-first ratio approaches deliver slightly higher expected return, slightly lower UP and slightly higher Roy's safety-first ratio than

the EURI. However, the EURI delivers much better EL and VaR than these two approaches. We conclude that our models are more than competitive with the benchmark approaches when compared across all the standard performance measures.

6.3. Sensitivity Analysis

It is of interest to potential users of our selection models to know when their relative advantage in performance over the benchmark approaches is greatest. Compared to other selection approaches, the difference with the EURI decreases with the target level. The reason for this is that as the target level increases, the project selections from the EURI, UP minimization, Markowitz model, and Roy's safety-first ratio maximization approaches become less risk averse. Consequently, the subsets of projects chosen become more similar to those chosen by the risk-neutral maximization of expected return approach. Hence the differences among the solutions from these models decrease.

The relative advantage of our models is also sensitive to the amount of interaction. We group the 200 project instances by the number of project bundles: low interaction (at most 62 bundles), medium interaction (from 63 through 66), and high interaction (at least 67). For each selection approach, Figure 1 shows the mean performance for each of the three groups, normalized by that of the EURI. For the VaR, we evaluate the performance using the maximum expected return as a reference point. The advantage of the EURI compared to Roy's safety-first ratio maximization approach, when measured by standard deviation, EL, CEL and VaR, increases with the density of interaction.

Figure 1. Performance profiles at various interaction densities.

Similar results apply to comparisons with the UP minimization approach. Therefore our recommendation to use the EURI over the other two approaches is stronger in project selection environments with more interactions.

Motivated by the example in Remark 1, we conduct a small computational test to study the importance of considering interactions. Our data set consists of 200 feasible instances that require the selection of 10 out of 20 available projects. The target for each instance is set at 80% of the maximal expected profit. Our results show that in 38% of the instances, the target is not achievable even in expectation if we ignore the interactions among the projects.

6.4. Robustness

We study the robustness of our EURI model by comparing two EURI solutions. The first uses full information about the distribution, whereas the second uses only knowledge of the bounded support and mean. We consider the same

200 instances as in §6.2, except for the uncertain factors. Under full information, the 50 uncertain factors follow beta distributions with parameters α_i, β_i , $i = 1, \dots, 50$, which are generated as $U(0.1, 0.9)$; under distributional ambiguity, we calculate the corresponding bound support $[0, 1]$ and mean support $\alpha_i/(\alpha_i + \beta_i)$ for each uncertain factor. We set $\varphi = 0.7$. For each project instance, we (a) find EURI solutions from the two information sets, (b) randomly generate a sample of size 100,000 for the 50 random factors following the beta distribution described above, and (c) compute the returns for both solutions from each information set.

Among the 200 project instances, there are 119 instances where the distributional ambiguity solution is the same as the full information solution. Hence we show average performance only over the remaining 81 instances, in Table 2.

The difference in performance between the two solutions is less than 1.5%, except for the EL criterion where it is 3.6%. We therefore conclude that the performance of our models is highly robust against distributional ambiguity.

Table 2. Robustness of EURI performance.

Distributional information	Criterion						
	Expected return	UP (%)	Standard deviation	EL	CEL	VaR @ 95%	VaR @ 99%
Full	49.07	4.422	7.024	0.1431	2.854	−37.42	−32.60
Robust	48.88	4.339	7.022	0.1477	2.887	−37.23	−32.38

Note. UP = underperformance probability; EL = expected loss; CEL = conditional expected loss; VaR = value-at-risk.

7. Concluding Remarks

This paper considers the problem of selecting projects when the return of each project is uncertain. Our modeling approach is general enough to allow us to consider interactions between different projects, and correlations between their uncertain returns. We describe a URI for this problem. Our model minimizes a special case, the EURI, which is the reciprocal of the ARA parameter, while keeping the certainty equivalent of the uncertain returns above a given target. The model is solved using a binary search on the ARA value of the project portfolio, with the solution of the subproblems by a Benders decomposition method. We demonstrate computationally that the EURI model identifies project portfolios which are more than competitive in achieving the target, compared to those found by classical approaches, including maximization of expected return, mean-variance analysis, minimization of UP, and Roy's safety-first ratio maximization. The EURI approach also provides robust performance in protecting against ambiguity in the data. For project selection problems that are constrained only by a budget, we describe a simple but highly accurate Heuristic EURI procedure.

The data requirements of our models are not excessive or unusual. We do not assume knowledge of a specific probability distribution for the factors that affect project return; instead, we only assume knowledge of bounded support and mean for the factor values. Covariance information is implied by common factors between projects, which, in practice, should be identified as a risk issue during preliminary evaluation by the project selection committee. Even if covariance information is not fully available, the robust selection model can still be used, based on partial covariance information. Finally, interaction effects between projects are routinely identified by the project selection committee during project definition.

Our results provide several insights that managers should find useful. First, it is now possible to design a EURI project portfolio that is least risky, subject to meeting a target certainty equivalent level. Second, this design can be achieved very accurately using a computationally efficient procedure. Third, the resulting project portfolios offer significant benefits over those obtained by the most widely used benchmark approaches from the literature. Fourth, it is possible to balance upside potential and downside risk accurately by adjusting the target level. Finally, in project selection situations that are constrained only by a budget, a simple spreadsheet-based procedure routinely provides almost exact EURI project portfolios.

Several opportunities exist for future research. First, in many practical projects, the initial investment cost is not predictable, and uncertainty about it can be incorporated into a EURI model. Second, a related extension is allowing the available budget to be random. In practice, available budgets for funding projects are often uncertain (Koç et al. 2009). Third, the EURI model should be applied to dynamic project selection problems. In such problems,

projects with random investment cost and return become available over time (Herbots et al. 2007). Consequently, some part of the available budget may need to be held in reserve for future opportunities. Fourth, the problem considered here can be generalized to allow for decisions about the timing of projects, to match resource requirements and resource availability over time. A EURI approach can be applied to this problem. Finally, it would be valuable to perform large-scale behavioral experiments on project selection, to determine the factors that influence how well the EURI explains those decisions in practice. We hope that our work will encourage future research in these interesting and practically important directions.

Supplemental Material

Supplemental material to this paper is available at <http://dx.doi.org/10.1287/opre.2015.1382>.

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